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3D horizontal memory cell with sequential 3D vertical stacking

Abstract

A semiconductor device includes a memory cell unit positioned over a substrate. The memory cell unit includes a transistor and a capacitor. The capacitor includes an inner conductor, a capacitor dielectric all around the inner conductor, an outer conductor all around the capacitor dielectric, and dielectric support structures below the inner conductor. The capacitor is elongated in a length direction parallel to a working surface of the substrate, and the dielectric support structures are spaced along the length direction. The transistor includes a channel structure, a gate structure all around the channel structure, and source/drain (S/D) regions on opposing ends of the channel structure.

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Background/Summary

INCORPORATION BY REFERENCE (1) This present disclosure claims the benefit of U.S. Provisional Application No. 63/320,457, filed on Mar. 16, 2022, which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) This disclosure relates to microelectronic devices including semiconductor devices, transistors, and integrated circuits, and methods of microfabrication.

BACKGROUND

(2) In the manufacture of a semiconductor device (especially on the microscopic scale), various fabrication processes are executed such as film-forming depositions, etch mask creation, patterning, material etching and removal, and doping treatments. These processes are performed repeatedly to form desired semiconductor device elements on a substrate. Historically, with microfabrication, transistors have been created in one plane, with wiring/metallization formed above the active device plane, and have thus been characterized as two-dimensional (2D) circuits or 2D fabrication. Scaling efforts have greatly increased the number of transistors per unit area in 2D circuits, yet scaling efforts are running into greater challenges as scaling enters single digit nanometer semiconductor device fabrication nodes. Semiconductor device fabricators have expressed a desire for three-dimensional (3D) semiconductor circuits in which transistors are stacked on of each other.

SUMMARY

(3) According to a first aspect of the disclosure, a semiconductor device is provided. The semiconductor device includes a memory cell unit positioned over a substrate. The memory cell unit includes a transistor and a capacitor. The capacitor includes an inner conductor, a capacitor dielectric all around the inner conductor, an outer conductor all around the capacitor dielectric, and dielectric support structures below the inner conductor. The capacitor is elongated in a length direction parallel to a working surface of the substrate, and the dielectric support structures are spaced along the length direction. The transistor includes a channel structure, a gate structure all around the channel structure, and source/drain (S/D) regions on opposing ends of the channel structure.

(4) In some embodiments, the capacitor dielectric and the outer conductor, below the inner conductor, are divided by the dielectric support structures and are discontinuous along the length direction. In some embodiments, the capacitor dielectric and the outer conductor, above the inner conductor, are undivided and are continuous along the length direction.

(5) In some embodiments, the dielectric support structures are in contact with the inner conductor.

(6) In some embodiments, the dielectric support structures are surrounded by the outer conductor below the inner conductor.

(7) In some embodiments, a plurality of the memory cell units are stacked over the substrate in a vertical direction perpendicular to the working surface of the substrate. In some embodiments, each capacitor dielectric and each outer conductor, below an uppermost inner conductor, are divided by respective dielectric support structures and are discontinuous along the length direction. An uppermost capacitor dielectric and an uppermost outer conductor, above the uppermost inner conductor, are undivided and are continuous along the length direction. In some embodiments, the plurality of the memory cell units includes a common gate structure for a plurality of transistors, and a common outer conductor for a plurality of capacitors.

(8) In some embodiments, the cell unit includes a semiconductor layer that includes a source region of the transistor, the channel structure of the transistor, a drain region of the transistor, and the inner conductor of the capacitor connected in series. In some embodiments, the semiconductor layer includes doped silicon. The channel structure includes a different dopant type from the source region, the drain region and the inner conductor.

(9) In some embodiments, the inner conductor, the capacitor dielectric, and the outer conductor are co-axial.

(10) In some embodiments, a plurality of the memory cell units are arranged in a width direction that is parallel to a working surface of the substrate and perpendicular to the length direction. The semiconductor device further includes isolation structures that separate a plurality of transistors of

the plurality of the memory cell units, and a common outer conductor for a plurality of capacitors of the plurality of the memory cell units.
(11) In some embodiments, the inner conductor includes a doped semiconductor. The capacitor dielectric includes a high-k dielectric. The outer conductor includes a metal.
(12) According to a second aspect of the disclosure, a method of manufacturing a semiconductor device is provided. The method includes forming a layer stack over a substrate. The layer stack is formed by forming a semiconductor layer over the substrate. Dielectric support structures are formed below and in contact with the semiconductor layer and spaced along a length direction parallel to a working surface of the substrate. A sacrificial material is formed over and below the semiconductor layer. The sacrificial material below the semiconductor layer is divided by the dielectric support structures. A capacitor is formed by removing the sacrificial material while the dielectric support structures remain in contact with the semiconductor layer. The capacitor is elongated in the length direction. A transistor is formed adjacent to the capacitor.
(13) In some embodiments, the dielectric support structures are configured to provide structural support to prevent collapse of the semiconductor layer when the sacrificial material is removed.
(14) In some embodiments, the semiconductor layer is doped such that the semiconductor layer includes source/drain regions and a channel structure of the transistor, and an inner conductor of the capacitor. In some embodiments, a portion of the sacrificial material in contact with the inner conductor is removed. A capacitor dielectric is formed all around the inner conductor. An outer conductor is formed all around the capacitor dielectric. In some embodiments, the capacitor dielectric and the outer conductor, below the inner conductor, are divided by the dielectric support structures and are discontinuous along the length direction. The capacitor dielectric and the outer conductor, above the inner conductor, are undivided and are continuous along the length direction. In some embodiments, a portion of the sacrificial material in contact with the channel structure is removed. A gate structure is formed all around the channel structure.
(15) In some embodiments, the semiconductor layer is divided in a width direction that is parallel to the working surface of the substrate and perpendicular to the length direction. A plurality of transistors are formed and spaced in the width direction. A plurality of capacitors are formed in the width direction.
(16) In some embodiments, the forming the layer stack further includes forming an additional semiconductor layer over the semiconductor layer. Additional dielectric support structures are formed below and in contact with the additional semiconductor layer and spaced along the length direction. The sacrificial material is formed over the additional semiconductor layer. The sacrificial material below the additional semiconductor layer is divided by the additional dielectric support structures.
(17) Note that this summary section does not specify every embodiment and/or incrementally novel aspect of the present disclosure or claimed invention. Instead, this summary only provides a preliminary discussion of different embodiments and corresponding points of novelty. For additional details and/or possible perspectives of the invention and embodiments, the reader is directed to the Detailed Description section and corresponding figures of the present disclosure as further discussed below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be increased or reduced for clarity of discussion.

(2) FIG. **1**A shows a top view of a semiconductor device in accordance with one embodiment of the present disclosure.
(3) FIGS. **1**B, **1**C and **1**D show vertical cross-sectional views taken along line cuts AA′, BB′ and CC′ in FIG. **1**A respectively in accordance with some embodiments of the present disclosure.
(4) FIG. **2**A shows a top view of a semiconductor device in accordance with another embodiment of the present disclosure.
(5) FIGS. **2**B, **2**C and **2**D show vertical cross-sectional views taken along line cuts DD′, EE′ and FF′ in FIG. **2**A respectively in accordance with some embodiments of the present disclosure.
(6) FIG. **3** shows a flow chart of a process for manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.
(7) FIGS. **4**A, **5**A, **6**A, **7**A, **8**A and **9**A show top views of a semiconductor device at various intermediate steps of manufacturing, in accordance with some embodiments of the present disclosure.
(8) FIGS. **4**B, **5**B, **5**C, **6**B, **6**C, **7**B, **7**C, **8**B, **9**B, **9**C and **9**D show vertical cross-sectional views of a semiconductor device at various intermediate steps of manufacturing, in accordance with some embodiments of the present disclosure.
(9) FIGS. **10**A, **11**A, **12**A, **13**A, **14**A, **15**A, **16**A and **17**A show top views of a semiconductor device at various intermediate steps of manufacturing, in accordance with some embodiments of the present disclosure.
(10) FIGS. **10**B, **11**B, **11**C, **12**B, **12**C, **13**B, **13**C, **14**B, **14**C, **15**B, **16**B, **16**C, **17**B, **17**C and **17**D show vertical cross-sectional views of a semiconductor device at various intermediate steps of manufacturing, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, spatially relative terms, such as “top,” “bottom,” “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(12) The order of discussion of the different steps as described herein has been presented for clarity sake. In general, these steps can be performed in any suitable order. Additionally, although each of the different features, techniques, configurations, etc. herein may be discussed in different places of this disclosure, it is intended that each of the concepts can be executed independently of each other or in combination with each other. Accordingly, the present invention can be embodied and viewed in many different ways.
(13) 3D integration, i.e. the vertical stacking of multiple devices, aims to overcome scaling limitations experienced in planar devices by increasing transistor density in volume rather than area. Although device stacking has been successfully demonstrated and implemented by the flash memory industry with the adoption of 3D NAND, application to random logic designs is

substantially more difficult. 3D integration for logic chips (CPU (central processing unit), GPU (graphics processing unit), FPGA (field programmable gate array, SoC (System on a chip)) is being pursued.

(14) Techniques herein integrate a novel 3D horizontal memory cell with sequential 3D vertical stacking. Any deposited material can be used for the channel regions. Example embodiments include making a DRAM cell that has one pass transistor and one capacitor that is horizontal integrated. A 3D stack of N tall DRAM cells is achieved herein. Another novel aspect is supporting a relatively large capacitor in the horizontal dimension by dielectric support pillars between adjacent 3D nanosheet capacitor build. Embodiments include an integration technique to make a co-axial cylinder capacitor for ultimate charge storage for DRAM memory cell. Examples show 3D nanosheet transistor and wrapped-around 3D capacitor with doped polysilicon. There are options for connections for source/drain, gate and capacitor shown with the new memory cell architecture herein.

(15) FIG. **1**A shows a top view of a semiconductor device **100** in accordance with one embodiment of the present disclosure. FIGS. **1**B, **1**C and **1**D show vertical cross-sectional views taken along line cuts AA′, BB′ and CC′ in FIG. **1**A respectively in accordance with some embodiments of the present disclosure.

(16) As illustrated, the semiconductor device **100** includes a stack of dynamic random access memory (DRAM) cell units (hereinafter referred to as a memory stack **110**) over a substrate **101** in the Z direction. The memory stack **110** includes a stack of capacitors (hereinafter referred to as a capacitor stack **112**) and a stack of transistors (hereinafter referred to as a transistor stack **113**) adjacent to each other. Consider a cell unit **111** (or a memory cell unit **111**) of the memory stack **110** for example. The cell unit **111** includes a capacitor **120** that is configured to be electrically coupled to a transistor **130** adjacent to the capacitor **120**.

(17) Specifically, the capacitor **120** includes an inner conductor **121**, a capacitor dielectric **123** (also shown as **123***a* and **123***b*) all around the inner conductor **121**, and an outer conductor **125** (also shown as **125***a* and **125***b*) all around the capacitor dielectric **123**. The capacitor **120** can also include dielectric support structures **129** that are spaced along the X direction and positioned below the inner conductor **121**. The dielectric support structures **129** can be in (direct) contact with the inner conductor **121** and thus configured to provide structural support (or physical support, or mechanical support) to prevent collapse of the inner conductor **121** during manufacturing, as will be explained in detail later.

(18) As shown in FIG. **1**D, the capacitor dielectric **123***b* and the outer conductor **125***b*, below the inner conductor **121** in the Z direction, are divided by the dielectric support structures **129** and are thus discontinuous along the X direction. The capacitor dielectric **123***a* and the outer conductor **125***a*, above the inner conductor **121** in the Z direction, are undivided and are thus continuous along the X direction. While not shown, in some embodiments, the dielectric support structures **129** are surrounded by the capacitor dielectric **123***b* and the outer conductor **125***b* below the inner conductor **121** in the XY plane. In addition, the dielectric support structures **129** can be rectangular or cylindrical pillars. In this case, the capacitor dielectric **123***b* and the outer conductor **125***b*, below the inner conductor **121** in the Z direction, can be undivided and continuous along the X direction in a different XZ cross-sectional view (not shown).

(19) In a non-limiting example, the inner conductor **121** includes a doped semiconductor material. The capacitor dielectric **123** includes a high-k dielectric, and the outer conductor **125** includes a metal material. Notably, the capacitor **120** is elongated in the X direction (relative to the Z direction). Dimensions and shape of the capacitor **120** (e.g. a length of the inner conductor **121** in the X direction and a thickness of the inner conductor **121** in the Z direction) may vary, depending on specific design requirements. For example, the inner conductor **121** can be a nanosheet while the capacitor dielectric **123** and the outer conductor **125** are disposed all around the nanosheet in the YZ plane. The inner conductor **121**, the capacitor dielectric **123** and the outer conductor **125**

can be co-axial. Additionally, the capacitor dielectric **123** may include extra portions that are disposed between the outer conductor **125***b* and the dielectric support structures **129**.

(20) The transistor **130** includes a channel structure **131** and a gate structure **132** all around the channel structure **131**. The gate structure **132** includes at least one gate dielectric **133** (such as a high-k dielectric) and at least one gate metal **134** (such as a work function metal (WFM)). The transistor **130** also includes S/D regions (e.g. **135** and **136**) positioned on two opposing ends of the channel structure **131**. Accordingly, the channel structure **131** can be configured to have a current flow path in the X direction. In this example, the transistor **130** includes a source region **136** and a drain region **135**.

(21) Further, the cell unit **111** can include a semiconductor layer **115** which includes the source region **136**, the channel structure **131**, the drain region **135** and the inner conductor **121** connected in series. In other words, the source region **136**, the channel structure **131**, the drain region **135** and the inner conductor **121** can be co-planar and configured to be electrically coupled. Therefore, the cell unit **111** can be configured to function as a horizontal cell and thus can be stacked in the Z direction. Particularly, the drain region **135** and the inner conductor **121** can be connected seamlessly or formed as a single piece.

(22) In some embodiments, the semiconductor layer **115** includes doped silicon. The channel structure **131** can include a different dopant type from the source region **136**, the drain region **135** and the inner conductor **121**. In one example, the channel structure **131** includes p-type polysilicon while the source region **136**, the drain region **135** and the inner conductor **121** include n-type polysilicon. In another example, the channel structure **131** includes n-type polysilicon while the source region **136**, the drain region **135** and the inner conductor **121** include p-type polysilicon.

(23) In some embodiments, the semiconductor device **100** can include dielectric materials, e.g. as shown by **103**, **105**, **123** and **133**. The dielectric materials may also be referred to as isolation structures, isolation layers, diffusion breaks, gate dielectrics, capping layers, etc. depending on functions thereof. For example in FIG. **1**B, the dielectric material **103** can separate transistor stacks (e.g. **113**) from each other and thus be referred to as isolation structures. The dielectric materials may be the same or different from each other.

(24) In the example of FIG. **1**, the memory stack **110** includes one cell unit **111** stacked in the Z direction. That is, the capacitor stack **112** includes one capacitor **120** stacked in the Z direction while the transistor stack **113** includes one transistor **130** stacked in the Z direction. It should be understood that the memory stack **110** can include a plurality of or any number of cell units **111** stacked in the Z direction. Moreover, other cell units can be similar to the cell unit **111** while having some differences. For example, channel structures of cell units can include different chemical compositions from one another. That is, the channel structures can include different semiconductor materials, different dopants and/or different dopant concentration profiles. Similarly, inner conductors of capacitors can also include different chemical compositions from one another.

(25) Further, the semiconductor device **100** can include any number of memory stacks **110** arranged in the Y direction in the XY plane. Isolation structures (e.g. **103**) can separate a plurality of transistors of the plurality of the memory cell units **111** from each other in the Y direction. The semiconductor device **100** can include a (common) outer conductor **125** for a plurality of capacitors of the plurality of the memory cell units **111**. In some embodiments, the isolation structures may extend further in the X direction to divide the (common) outer conductor **125** into independent outer conductors (not shown); that is, the plurality of capacitors can be separated in the Y direction by one or more dielectric materials.

(26) FIG. **2**A shows a top view of a semiconductor device **200** in accordance with another embodiment of the present disclosure. FIGS. **2**B, **2**C and **2**D show vertical cross-sectional views taken along line cuts DD′, EE′ and FF′ in FIG. **2**A respectively in accordance with some embodiments of the present disclosure. As the embodiment of the semiconductor device **200** is

similar to the embodiment of the semiconductor device **100**, descriptions herein will be provided with emphasis placed on difference. Particularly, similar or identical components are labeled with similar or identical numerals unless specified otherwise.

(27) As illustrated, the semiconductor device **200** can include two cell units **111** stacked in the Z direction. Accordingly, the memory stack **110** includes a capacitor **120***a* stacked over a capacitor **120***b*. Similarly, the capacitor stack **112** includes inner conductors **121***a* and **121***b*, the capacitor dielectric **123** (also shown as **123***a*, **123***b* and **123***c*) and the outer conductor **125** (also shown as **125***a*, **125***b* and **125***c*). The capacitor stack **112** can also include dielectric support structures **129***a* and **129***b*, which correspond to the dielectric support structures **129**.

(28) In the examples of FIGS. **2**A-**2**D, the capacitor dielectric **123***a* and the outer conductor **125***a*, above the inner conductor **121***a* in the Z direction, are undivided and are thus continuous along the X direction. The capacitor dielectric **123***b* and the outer conductor **125***b*, between the inner conductor **121***a* and the inner conductor **121***b* in the Z direction, are divided by the dielectric support structures **129***a* and are thus discontinuous along the X direction. The capacitor dielectric **123***c* and the outer conductor **125***c*, below the inner conductor **121***b* in the Z direction, are divided by the dielectric support structures **129***b* and are thus discontinuous along the X direction. Additionally, the outer conductor **125** can be a common outer conductor for a plurality of capacitors arranged in the XY plane as well as stacked in the Z direction.

(29) As can be understood, in other embodiments (not shown), the semiconductor device **200** can include any number of cell units **111** stacked in the Z direction and thus any number of capacitors stacked in the Z direction. Accordingly, each capacitor dielectric (e.g. **123***b* and **123***c*) and each outer conductor (**125***b* and **125***c*), below an uppermost inner conductor (e.g. **121***a*), are divided by respective dielectric support structures (e.g. **129***a* and **129***b*) and are discontinuous along the X direction. An uppermost capacitor dielectric (e.g. **123***a*) and an uppermost outer conductor (e.g. **125***a*), above the uppermost inner conductor (e.g. **121***a*), are undivided and are continuous along the X direction.

(30) Still referring to FIGS. **2**A-**2**D, the transistor stack **113** can include two transistors **130** stacked in the Z direction. The gate structure **132** can be a common gate structure for two transistors **130**.

(31) FIG. **3** shows a flow chart of a process **300** for manufacturing a semiconductor device, such as the semiconductor device **100**, the semiconductor device **200** or the like, in accordance with some embodiments of the present disclosure. The process **300** starts with Step S**210** where a layer stack is formed over a substrate. Particularly, a semiconductor layer can be formed over the substrate. Dielectric support structures can be formed below and in contact with the semiconductor layer and spaced along a length direction parallel to a working surface of the substrate. A sacrificial material can be formed over and below the semiconductor layer. The sacrificial material below the semiconductor layer is divided by the dielectric support structures.

(32) The process **300** then proceeds to Step S**220** where a capacitor is formed by removing the sacrificial material while the dielectric support structures remain in contact with the semiconductor layer. The capacitor is elongated in the length direction. The process **300** then proceeds to Step S**220** by forming a transistor adjacent to the capacitor.

(33) FIGS. **4**A, **5**A, **6**A, **7**A, **8**A and **9**A show top views of a semiconductor device **400** at various intermediate steps of manufacturing, in accordance with some embodiments of the present disclosure. For example, FIGS. **4**A-**9**A can show horizontal DRAM access with high density nanosheet transistor and wrapped-around capacitor with doped polysilicon. There are options for connections for source/drain, gate and capacitor. Such an embodiment is highly suitable for hierarchical design of N-number stack.

(34) FIG. **4**B can show a vertical cross-sectional view taken along the line cut GG′ in FIG. **4**A. In FIGS. **4**A-**4**B, a layer stack **350** is formed over a substrate **301**. Specifically, the layer stack **350** includes a semiconductor layer **315** formed over the substrate **301**, dielectric support structures **329** and a sacrificial material **351** (e.g. shown as **351***a*, **351***a*′, **351***b* and **351***b*′). The dielectric support

structures **329** are formed below and in contact with the semiconductor layer **351** and spaced along the X direction. The sacrificial material **351** is formed over (e.g. **351***a* and **351***a*′) and below (e.g. **351***b* and **351***b*′) the semiconductor layer **315**. The semiconductor layer **315** can include a source region **336**, a channel structure **331** and a drain region **335** in a transistor region **341** and an inner conductor **321** in a capacitor region **342**. Additionally, the sacrificial material **351***b* below the semiconductor layer **315** can be divided by the dielectric support structures **329**. For example, the sacrificial material **351***b* can be thin slices while the semiconductor layer **315** can be divided into separate or individual lines.

(35) In some embodiments, in order to form the layer stack **350**, a layer of a first dielectric **303** is deposited on silicon (e.g. **301**) or another substrate. Using a bottom DRAM gate access mask, the layer of the first dielectric **303** is directionally etched—partially—while keeping insulation with Si substrate (e.g. **301**). After stripping off a photoresist, the sacrificial material **351***b* and **351***b*′ (e.g. a second dielectric) is deposit-filled and planarized, for example by chemical-mechanical polishing (CMP). The capacitor region **354** will be used for forming at least one capacitor which is a long and thin strip. Therefore, it needs some support to bolster its physical stability. Although, it will slightly disturb the capacitance value, the capacitance value would still be in acceptable range.

(36) Then, p-doped polysilicon (e.g. **315**) that is relatively lightly doped is deposited before being directionally doped by ion-implantation with n+ dopants using a blocking mask. As a result, an n-doped polysilicon (e.g. **331**) is formed and has a different dopant type from other portions of the original p-doped polysilicon (e.g. **315**). The source region **336** can be inputs for DRAM, while the drain region **335** and the inner conductor **321** are shorted together. Subsequently, the sacrificial material is deposited before lithography is executed with an isolation DRAM mask to make the sacrificial material discontinuous (e.g. **351***a* and **351***a*′). Next, after stripping off a photoresist, the first dielectric **303** is deposit-filled and planarized by CMP.

(37) Note that the substrate **301** can correspond to the substrate **101**. The semiconductor layer **315** can correspond to the semiconductor layer **115**. The dielectric support structures **329** can correspond to the dielectric support structures **129**. The source region **336** can correspond to the source region **136**. The channel structure **331** can correspond to the channel structure **131**. The drain region **335** can correspond to the drain region **135**. The inner conductor **321** can correspond to the inner conductor **121**.

(38) FIGS. **5**B and **5**C can respectively show vertical cross-sectional views taken along line cuts HH′ and II′ in FIG. **5**A. In FIGS. **5**A-**5**C, openings **353** are formed to divide the channel structure **331** into independent channel structure spaced in the Y direction. Through the openings **353**, the sacrificial material **351***a*′ and **351***b*′, which is in contact with the channel structure **331** in the transistor region **341**, is removed. As a result, the channel structure **331** is exposed.

(39) In some embodiments, a hard mask layer of a third dielectric (e.g. **361**) is deposited. Then, lithography is executed with a DRAM gate mask to directionally etch the third dielectric (e.g. **361**), the first dielectric **303**, the sacrificial material **315***a*′, polysilicon (e.g. **331**) and the sacrificial material **315***b*′. Etch can stop anywhere after passing polysilicon (e.g. **331**) or etch stop on Si (e.g. **301**). After stripping a photoresist, the sacrificial material **315***a*′ and **315***b*′ is removed by selective etching. This will create space for gate wrapping around p-doped poly.

(40) FIGS. **6**B and **6**C can respectively show vertical cross-sectional views taken along line cuts JJ′ and KK′ in FIG. **6**A. In FIGS. **6**A-**6**C, a gate structure **332** is formed all around the channel structure **331**. As a result, a transistor **330** is formed. The gate structure **332** includes at least one gate dielectric **333** (such as a high-k dielectric) and at least one gate metal **334** (such as a work function metal (WFM)). For example, a thin high-k dielectric (e.g. **333**) can be deposited by atomic layer deposition, and a metal can be deposit-filled and planarized by CMP.

(41) Note that the transistor **330** can correspond to the transistor **130**. The gate structure **332** can correspond to the gate structure **132**. The at least one gate dielectric **333** can correspond to the at least one gate dielectric **133**. The at least one gate metal **334** can correspond to the at least one gate

metal **134**.

(42) FIGS. **7**B and **7**C can respectively show vertical cross-sectional views taken along line cuts LL′ and MM′ in FIG. **7**A. In FIGS. **7**A-**7**C, openings **355** are formed to divide the inner conductor **321** into independent inner conductors spaced in the Y direction. Through the openings **355**, the sacrificial material **351***a* and **351***b*, which is in contact with the inner conductor **321** in the capacitor region **342**, is removed. As a result, the inner conductor **321** is exposed. Note that the dielectric support structures **329** can be configured to provide structural support (or physical support, or mechanical support) to prevent collapse of the inner conductor **321** when the sacrificial material **351***a* and **351***b* is removed.

(43) In some embodiments, similar to FIGS. **5**A-**5**C, lithography is executed with a DRAM capacitor mask to etch directionally a hard mask layer of a fourth dielectric (e.g. **363**), the first dielectric **303**, the sacrificial material **351***a*, polysilicon (e.g. **331**), the sacrificial material **351***b* and the dielectric support structures **329**. Etch can stop anywhere after passing polysilicon (e.g. **321**) or etch stop on Si (e.g. **301**). Then, a photoresist is stripped off, and the sacrificial material **351***a* and **351***b* can be removed by selective etching. The pillars (e.g. **329**) can support a thin and long n-type polysilicon (e.g. **321**).

(44) FIG. **8**B can show a vertical cross-sectional view taken along the line cut NN′ in FIG. **8**A. In FIGS. **8**A-**8**B, a capacitor dielectric **323** (also shown as **323***a* and **323***b*) is formed all around the inner conductor **321**, and an outer conductor **325** (also shown as **325***a* and **325***b*) is formed all around the capacitor dielectric **323**. As a result, a capacitor **320** is formed, and a cell unit **311** (or a memory cell unit) is formed. For example, a high-k dielectric (e.g. **323**) can be formed by ALD, and a metal (e.g. **325**) can be deposited and planarized by CMP. Different high-k dielectric and/or metal can also be used as per requirement.

(45) Note that the cell unit **311** can correspond to the cell unit **111**. The capacitor **320** can correspond to the capacitor **120**. The capacitor dielectric **323**, **323***a* and **323***b* can respectively correspond to the capacitor dielectric **123**, **123***a* and **123***b*. The outer conductor **325**, **325***a* and **325***b* can respectively correspond to the outer conductor **125**, **125***a* and **125***b*.

(46) FIGS. **9**B, **9**C and **9**D can respectively show vertical cross-sectional views taken along line cuts PP′, QQ′ and RR′ in FIG. **9**A. In FIGS. **9**A-**9**D, isolation structures (e.g. **303**) are formed between transistors **330** arranged in the Y direction. For example, lithography can be executed with a slicing mask to directionally etch materials until the end of the first dielectric (e.g. **303**) or etch stop at Si (e.g. **301**). After stripping off a photoresist, the first dielectric (e.g. **303**) is deposit-filled and then planarized by CMP. As a result, the semiconductor device **400** can become the semiconductor device **100**. While not shown, any number of cell units **311** can be stacked in the Z direction.

(47) FIGS. **10**A, **11**A, **12**A, **13**A, **14**A, **15**A, **16**A and **17**A show top views of a semiconductor device **500** at various intermediate steps of manufacturing, in accordance with some embodiments of the present disclosure. In the examples of FIGS. **4**A-**9**A, when slicing, all other remaining doped Si (e.g. **315** in FIGS. **9**A-**9**D) has been separated. FIGS. **10**A-**17**A can show an alternative embodiment which is to slice the doped polysilicon early and then build the device. Besides that, FIGS. **10**A-**17**A includes a hierarchy concept of N-number of DRAM access.

(48) FIG. **10**B can show a vertical cross-sectional view taken along the line cut SS' in FIG. **10**A. In FIGS. **10**A-**10**B, a layer stack **350**′ is formed over the substrate **301**. Specifically, the layer stack **350**′ includes a semiconductor layer **315***b* formed over the substrate **301**, a semiconductor layer **315***a* formed over the semiconductor layer **315***b*, dielectric support structures **329***a* and **329***b*, and the sacrificial material **351** (e.g. shown as **351***a*, **351***a*′, **351***b*, **351***b*′, **351***c* and **351***c*′). The dielectric support structures **329***a* are formed below and in contact with the semiconductor layer **315***a* and spaced along the X direction. The dielectric support structures **329***b* are formed below and in contact with the semiconductor layer **315***b* and spaced along the X direction. The sacrificial material **351** is formed over (e.g. **351***a* and **351***a*′) and below (e.g. **351***b* and **351***b*′) the

semiconductor layer **315***a* as well as formed over (e.g. **351***b* and **351***b*′) and below (e.g. **351***c* and **351***c*′) the semiconductor layer **315***b*. The semiconductor layer **315***b* (or **315***a*) can include the source region **336**, the channel structure **331** and the drain region **335** in the transistor region **341** and the inner conductor **321** in the capacitor region **342**. Additionally, the sacrificial material **351***b* below the semiconductor layer **315***a* can be divided by the dielectric support structures **329***a* while the sacrificial material **351***c* below the semiconductor layer **315***b* can be divided by the dielectric support structures **329***b*.
(49) In some embodiments, the layer stack **350**′ herein can be formed in a similar way to the layer stack **350** in FIGS. **4**A-**4**B. That is, the sacrificial material **351***c* and **351***c*′ and the dielectric support structures **329***b* can be formed similarly to the sacrificial material **351***b* and **351***b*′ and the dielectric support structures **329** respectively. Likewise, the semiconductor layers **315***a* and **315***b* can be formed similarly to the semiconductor layer **315** by directional doping.
(50) Note that the semiconductor layers **315***a* and **315***b* can each correspond to the semiconductor layer **315**. The dielectric support structures **329***a* and **329***b* can correspond to the dielectric support structures **329**. The sacrificial material **351***c* and **351***c*′ can correspond to the sacrificial material **351***b* and **351***b*′.
(51) FIGS. **11**B and **11**C can respectively show vertical cross-sectional views taken along line cuts TT′ and UU′ in FIG. **11**A. In FIGS. **11**A-**11**C, the layer stack **350**′ is divided into separate or independent layer stacks spaced in the Y direction, for example by using an early slicing DRAM mask to etch through layers of the layer stack **350**′ with etch stop at Si (e.g. **301**) or the first dielectric **303**. Note that this step is different from the examples of FIGS. **4**A-**9**A as explained above.
(52) FIGS. **12**B and **12**C can respectively show vertical cross-sectional views taken along line cuts VV′ and WW′ in FIG. **12**A. In FIGS. **12**A-**12**C, a photoresist **365** is stripped off, and the first dielectric **303** is deposit-filled and then planarized by CMP.
(53) FIGS. **13**B and **13**C can respectively show vertical cross-sectional views taken along line cuts XX′ and YY′ in FIG. **13**A. In FIGS. **13**A-**13**C, the first dielectric **303** in the transistor region **341** is removed before the sacrificial material **351***a*′, **351***b*′ and **351***c*′ in the transistor region **341** is selectively removed. For example, a hard mask **367** (e.g. a dielectric) can be formed, and a DRAM gate mask can be used to directionally etch the first dielectric **303** in the transistor region **341**. After stripping off a photoresist, the third dielectric (e.g. **351***a*′, **351***b*′ and **351***c*′) can be removed from the transistor region **341** by selective etching.
(54) FIGS. **14**B and **14**C can respectively show vertical cross-sectional views taken along line cuts AAAA′ and BBBB′ in FIG. **14**A. In FIGS. **14**A-**14**C, the gate structure **332** is formed all around the channel structure **331**. As a result, transistors **330** are formed. The gate structure **332** includes the at least one gate dielectric **333** and the at least one gate metal **334**. For example, a thin high-k dielectric (e.g. **333**) can be deposited by atomic layer deposition, and a metal can be deposit-filled and planarized by CMP.
(55) FIG. **15**B can show a vertical cross-sectional view taken along the line cut CCCC′ in FIG. **15**A. In FIGS. **15**A-**15**B, the inner conductors **321***a* and **321***b* are divided into independent inner conductors spaced in the Y direction. The sacrificial material **351***a*, **351***b* and **351***c*, which is in contact with the inner conductors **321***a* and **321***b* in the capacitor region **342**, is removed. As a result, the inner conductors **321***a* and **321***b* are exposed. Note that the dielectric support structures **329***a* and **329***b* can be configured to provide structural support (or physical support, or mechanical support) to prevent collapse of the inner conductors **321***a* and **321***b* when the sacrificial material **351***a*, **351***b* and **351***c* is removed.
(56) In some embodiments, similar to FIGS. **7**A-**7**C, lithography can be executed using a DRAM capacitor mask (e.g. **369**) to directionally etch the first dielectric **303**, the sacrificial material **351***a*, the inner conductor **321***a*, the sacrificial material **351***b* and the dielectric support structures **329***a*, the inner conductor **321***b*, and the sacrificial material **351***c* and the dielectric support structures

329*b*. Then, the third dielectric (e.g. **351***a*, **351***b* and **351***c*) can be removed from the capacitor region **342** by selective etching.
(57) FIGS. **16**B and **16**C can respectively show vertical cross-sectional views taken along line cuts DDDD′ and EEEE′ in FIG. **16**A. In FIGS. **16**A-**16**C, the capacitor dielectric **323** (also shown as **323***a*, **323***b* and **323***c*) is formed all around the inner conductors **321***a* and **321***b*, and the outer conductor **325** (also shown as **325***a*, **325***b* and **325***c*) is formed all around the capacitor dielectric **323**. As a result, capacitors **320***a* and **320***b* are formed, and cell units **311** are formed. For example, a high-k dielectric (e.g. **323**) can be formed by ALD, and a metal (e.g. **325**) can be deposited and planarized by CMP. Different high-k dielectric and/or metal can also be used as per requirement. Note that the capacitor dielectric **323***c* can correspond to the capacitor dielectric **123***b*. The outer conductor **325***c* can correspond to the outer conductor **125***b*.
(58) FIGS. **17**B, **17**C and **17**D can respectively show vertical cross-sectional views taken along line cuts FFFF′, GGGG′ and HHHH′ in FIG. **17**A. In FIGS. **17**A-**17**D, isolation structures (e.g. **303**) are formed between transistors **330** arranged in the Y direction. As a result, the semiconductor device **500** can become the semiconductor device **200**. It should be understood that any number of cell units **311** can be stacked in the Z direction. For example, lithography can be executed with a slicing mask to directionally etch materials until the end of the first dielectric **303** or etch stop at Si (e.g. **301**). After stripping off a photoresist, the first dielectric **303** is deposit-filled and then planarized by CMP. Note that the slicing mask herein is different from that of FIGS. **9**A-**9**D. Because an early DRAM mask has been used to etch all the way in FIGS. **11**A-**11**C, this step is only slicing nanosheet transistors (e.g. **330**) without touching (or etching) the capacitors (e.g. **320***a* and **320***b*).
(59) In the preceding description, specific details have been set forth, such as a particular geometry of a processing system and descriptions of various components and processes used therein. It should be understood, however, that techniques herein may be practiced in other embodiments that depart from these specific details, and that such details are for purposes of explanation and not limitation. Embodiments disclosed herein have been described with reference to the accompanying drawings. Similarly, for purposes of explanation, specific numbers, materials, and configurations have been set forth in order to provide a thorough understanding. Nevertheless, embodiments may be practiced without such specific details. Components having substantially the same functional constructions are denoted by like reference characters, and thus any redundant descriptions may be omitted.
(60) Various techniques have been described as multiple discrete operations to assist in understanding the various embodiments. The order of description should not be construed as to imply that these operations are necessarily order dependent. Indeed, these operations need not be performed in the order of presentation. Operations described may be performed in a different order than the described embodiment. Various additional operations may be performed and/or described operations may be omitted in additional embodiments.
(61) “Substrate” or “wafer” as used herein generically refers to an object being processed in accordance with the invention. The substrate may include any material portion or structure of a device, particularly a semiconductor or other electronics device, and may, for example, be a base substrate structure, such as a semiconductor wafer, reticle, or a layer on or overlying a base substrate structure such as a thin film Thus, substrate is not limited to any particular base structure, underlying layer or overlying layer, patterned or un-patterned, but rather, is contemplated to include any such layer or base structure, and any combination of layers and/or base structures. The description may reference particular types of substrates, but this is for illustrative purposes only.
(62) The substrate can be any suitable substrate, such as a silicon (Si) substrate, a germanium (Ge) substrate, a silicon-germanium (SiGe) substrate, and/or a silicon-on-insulator (SOI) substrate. The substrate may include a semiconductor material, for example, a Group IV semiconductor, a Group III-V compound semiconductor, or a Group II-VI oxide semiconductor. The Group IV semiconductor may include Si, Ge, or SiGe. The substrate may be a bulk wafer or an epitaxial

layer.
(63) Those skilled in the art will also understand that there can be many variations made to the operations of the techniques explained above while still achieving the same objectives of the invention. Such variations are intended to be covered by the scope of this disclosure. As such, the foregoing descriptions of embodiments of the invention are not intended to be limiting. Rather, any limitations to embodiments of the invention are presented in the following claims.

Claims

1. A semiconductor device, comprising: a memory cell unit positioned over a substrate, wherein the memory cell unit comprises a transistor and a capacitor, wherein the capacitor comprises: an inner conductor, a capacitor dielectric all around the inner conductor, an outer conductor all around the capacitor dielectric, and dielectric support structures below the inner conductor, where the capacitor is elongated in a length direction parallel to a working surface of the substrate, and the dielectric support structures are spaced along the length direction, wherein the transistor comprises: a channel structure, a gate structure all around the channel structure, and source/drain (S/D) regions on opposing ends of the channel structure.
2. The semiconductor device of claim 1, wherein: the capacitor dielectric and the outer conductor, below the inner conductor, are divided by the dielectric support structures and are discontinuous along the length direction.
3. The semiconductor device of claim 2, wherein: the capacitor dielectric and the outer conductor, above the inner conductor, are undivided and are continuous along the length direction.
4. The semiconductor device of claim 1, wherein: the dielectric support structures are in contact with the inner conductor.
5. The semiconductor device of claim 1, wherein: the dielectric support structures are surrounded by the outer conductor below the inner conductor.
6. The semiconductor device of claim 1, wherein: a plurality of the memory cell units are stacked over the substrate in a vertical direction perpendicular to the working surface of the substrate.
7. The semiconductor device of claim 6, wherein: each capacitor dielectric and each outer conductor, below an uppermost inner conductor, are divided by respective dielectric support structures and are discontinuous along the length direction, and an uppermost capacitor dielectric and an uppermost outer conductor, above the uppermost inner conductor, are undivided and are continuous along the length direction.
8. The semiconductor device of claim 6, wherein the plurality of the memory cell units comprises: a common gate structure for a plurality of transistors; and a common outer conductor for a plurality of capacitors.
9. The semiconductor device of claim 1, wherein: the memory cell unit comprises a semiconductor layer that includes a source region of the transistor, the channel structure of the transistor, a drain region of the transistor, and the inner conductor of the capacitor connected in series.
10. The semiconductor device of claim 9, wherein: the semiconductor layer comprises doped silicon, and the channel structure includes a different dopant type from the source region, the drain region and the inner conductor.
11. The semiconductor device of claim 1, wherein: the inner conductor, the capacitor dielectric, and the outer conductor are co-axial.
12. The semiconductor device of claim 1, further comprising: a plurality of the memory cell units arranged in a width direction that is parallel to a working surface of the substrate and perpendicular to the length direction; isolation structures separating a plurality of transistors of the plurality of the memory cell units; and a common outer conductor for a plurality of capacitors of the plurality of the memory cell units.
13. A method of manufacturing a semiconductor device, the method comprising: forming a layer

stack over a substrate by: forming a semiconductor layer over the substrate, forming dielectric support structures that are below and in contact with the semiconductor layer and spaced along a length direction parallel to a working surface of the substrate, and forming a sacrificial material over and below the semiconductor layer, wherein the sacrificial material below the semiconductor layer is divided by the dielectric support structures; forming a capacitor by removing the sacrificial material while the dielectric support structures remain in contact with the semiconductor layer, wherein the capacitor is elongated in the length direction; and forming a transistor adjacent to the capacitor.

14. The method of claim 13, wherein: the dielectric support structures are configured to provide structural support to prevent collapse of the semiconductor layer when the sacrificial material is removed.

15. The method of claim 13, further comprising: doping the semiconductor layer such that the semiconductor layer comprises source/drain regions and a channel structure of the transistor, and an inner conductor of the capacitor.

16. The method of claim 15, further comprising: removing a portion of the sacrificial material in contact with the inner conductor; forming a capacitor dielectric all around the inner conductor; and forming an outer conductor all around the capacitor dielectric.

17. The method of claim 16, wherein: the capacitor dielectric and the outer conductor, below the inner conductor, are divided by the dielectric support structures and are discontinuous along the length direction, and the capacitor dielectric and the outer conductor, above the inner conductor, are undivided and are continuous along the length direction.

18. The method of claim 15, further comprising: removing a portion of the sacrificial material in contact with the channel structure; and forming a gate structure all around the channel structure.

19. The method of claim 13, further comprising: dividing the semiconductor layer in a width direction that is parallel to the working surface of the substrate and perpendicular to the length direction; forming a plurality of transistors spaced in the width direction; and forming a plurality of capacitors in the width direction.

20. The method of claim 13, wherein the forming the layer stack further comprises: forming an additional semiconductor layer over the semiconductor layer; forming additional dielectric support structures below and in contact with the additional semiconductor layer and spaced along the length direction; and forming the sacrificial material over the additional semiconductor layer, wherein the sacrificial material below the additional semiconductor layer is divided by the additional dielectric support structures.